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AA216 : Flight Mechanics and Classical Control

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Problem Statement: PID Control of Quadrotor Altitude Stabilization Under Wind Gust Disturbances

- **Type of Control System:** MIMO (Multi-Input Multi-Output) Feedback Control System with Disturbance Rejection.
- **Problem Description:** A quadrotor drone experiences altitude instability due to varying wind gusts, sensor noise, and nonlinear motor dynamics. The flight controller must track a desired altitude reference while simultaneously rejecting persistent external wind disturbances in the range of 0–15 m/s. **Without active control, the open-loop plant is unstable** – its second-order airframe dynamics contain a pair of right-half-plane (RHP) poles that produce divergent oscillations and exponential growth, which can structurally stress the frame and degrade mission performance.
- **Control Challenge:** The open-loop plant contains an **unstable conjugate pole pair** ($\zeta = -0.4$) that causes exponentially growing oscillatory behaviour. Wind gusts act as a persistent step-like disturbance input, demanding integral action to drive steady-state error to zero. Sensor noise, actuator bandwidth limitations (modelled by motor dynamics), and the requirement for smooth ascent/descent make this a multi-objective design problem requiring a carefully tuned PID controller to **pull the unstable poles into the LHP**.
- **Approach:** A classical PID controller is designed and analysed using BIBO stability theory, Routh–Hurwitz stability criterion, Root Locus method, Bode frequency-response analysis, and steady-state error calculations. MATLAB/Simulink is used for simulation and verification.

Control Objective

Towards Stability

- Ensure bounded altitude response despite sustained wind disturbances of up to 15 m/s.
- Prevent divergent oscillatory behaviour that would mechanically damage the quadrotor frame.
- Maintain closed-loop stability across the full operating wind-speed envelope by keeping **all closed-loop poles strictly in the left-half s-plane** (stabilizing the initially unstable plant).

Towards Performance

- Achieve less than 5% steady-state error in altitude tracking for a step reference input.
- Attain a rise time of less than 2 s for a unit-step altitude command.
- Limit overshoot to ± 0.5 m from the commanded altitude.
- Reject wind-gust disturbances with a settling time of less than 3 s.

Objective Summary

The primary objective is to design a PID controller $C(s) = K_p + \frac{K_i}{s} + K_d s$ that **renders the initially unstable closed-loop system stable**, rejects persistent wind disturbances (via integral action), and meets all four quantitative performance specifications simultaneously.

System Block Diagram

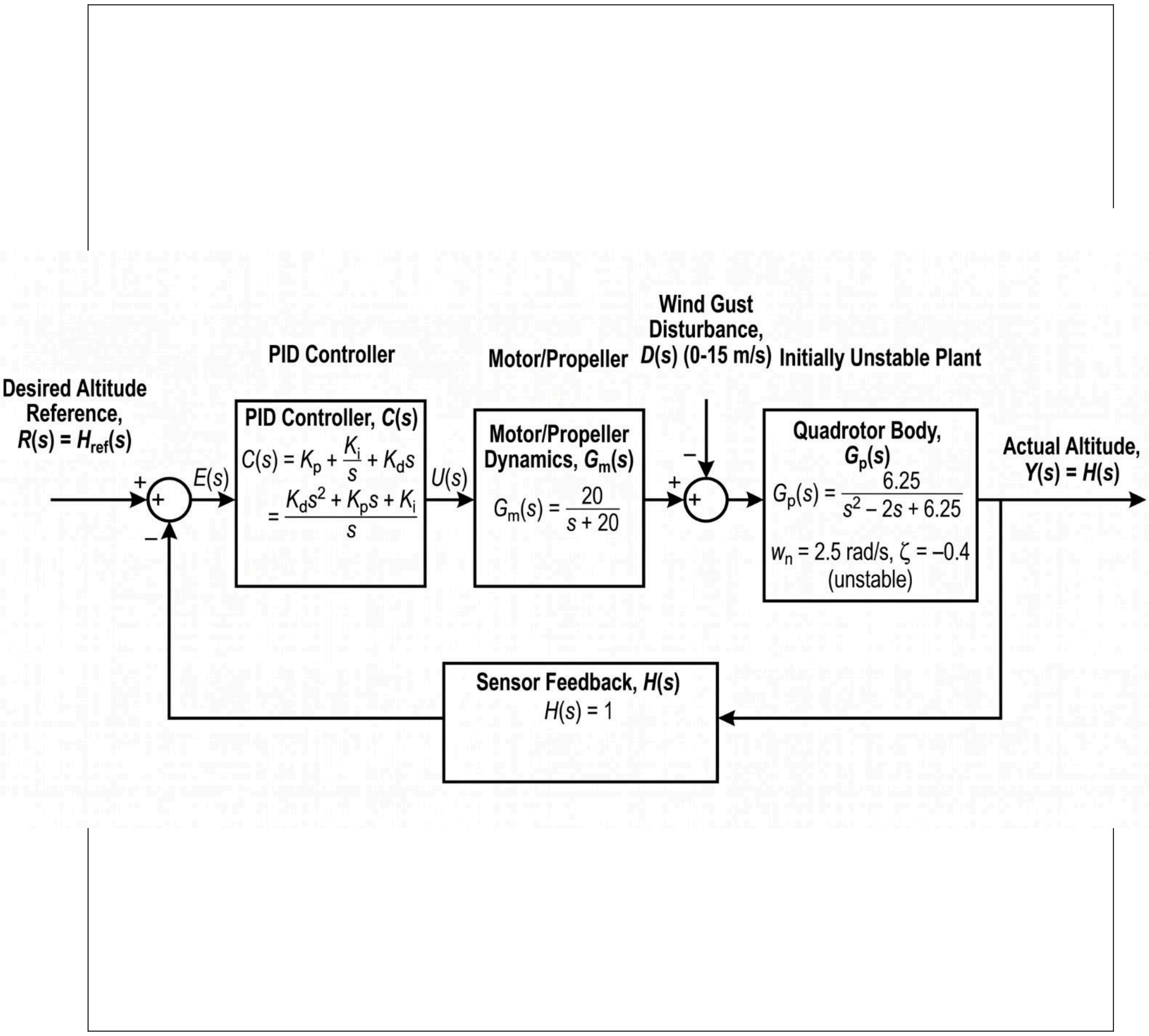

Figure 1: Closed-loop block diagram of the quadrotor altitude control system.

Mathematical Model
Plant Transfer Function (Quadrotor body):

$$G_p(s) = \frac{\omega_n^2}{s^2 - 2\zeta\omega_n s + \omega_n^2} = \frac{6.25}{s^2 - 2s + 6.25} \tag{1}$$

where $\omega_n = 2.5$ rad/s and $\zeta = -0.4$ (**unstable, negative damping**).
Motor / Propeller Dynamics:

$$G_m(s) = \frac{20}{s + 20} \tag{2}$$

Full Open-Loop Transfer Function:

$$G_{OL}(s) = G_m(s)G_p(s) = \frac{20 \times 6.25}{(s + 20)(s^2 - 2s + 6.25)} = \frac{125}{(s + 20)(s^2 - 2s + 6.25)} \tag{3}$$

PID Controller:

$$C(s) = K_p + \frac{K_i}{s} + K_d s = \frac{K_d s^2 + K_p s + K_i}{s} \tag{4}$$

System Characteristics

- **Open-Loop Poles:**

$$
\begin{aligned}
p_1 &= -20 \quad \text{(motor pole — fast, well-damped)} \\
p_{2,3} &= +1 \pm j\,2.179 \quad \text{(quadrotor poles — } \textbf{unstable}\text{, } \zeta = -0.4\text{)}
\end{aligned}
$$

- **Finite Zeros:** None.
- **Open-Loop Stability (BIBO):** Two poles lie in the open right-half s-plane $\Rightarrow$ the open-loop plant is **NOT BIBO stable (unstable)**.
- **Concern:** The unstable poles at $p_{2,3}$ produce divergent oscillations and exponential growth in open-loop step response, necessitating closed-loop PID control with proportional and derivative stabilization.
- **Steady-State Error (open-loop, type-0):** A step disturbance produces unbounded error due to instability, motivating the use of PID structure.

Project Report

1. Plant Transfer Function & Controller Design

The complete open-loop plant is given by Equation (3). A PID controller (Equation (4)) is selected because:

- **Integral (K_i):** Eliminates steady-state error due to step wind disturbances (Type-1 closed-loop system).
- **Proportional (K_p):** Provides primary feedback gain to pull the unstable RHP poles into the LHP.
- **Derivative (K_d):** Adds phase lead to damp the oscillatory unstable poles and reduce overshoot.

The closed-loop transfer function with unity sensor feedback $H(s) = 1$ is:

$$T(s) = \frac{C(s)G_m(s)G_p(s)}{1 + C(s)G_m(s)G_p(s)} \tag{5}$$

2. Stability Analysis

(a) BIBO Stability

The open-loop plant is **unstable** (two RHP poles). Closed-loop stability is verified by confirming that all roots of the characteristic equation $1 + C(s)G_m(s)G_p(s) = 0$ lie in the LHP after PID compensation.

(b) Routh–Hurwitz Criterion

The closed-loop characteristic polynomial (with PID controller) is obtained by expanding the denominator of $T(s)$.

$$\Delta(s) = s^4 + 18s^3 + (-33.75 + 125K_d)s^2 + (125 + 125K_p)s + 125K_i \tag{6}$$

For the chosen gains $K_p = 3.0,\ K_i = 1.5,\ K_d = 0.8$, the characteristic polynomial is:

$$\Delta(s) = s^4 + 18s^3 + 66.25\,s^2 + 500\,s + 187.5 \tag{7}$$

Routh Array:

s^4	1	66.25	187.5
s^3	18	500	0
s^2	$66.25 - \frac{500}{18} = 38.47$	187.5	
s^1	$500 - \frac{18\times187.5}{38.47} = 412.3$	0	
s^0	187.5		

All elements in the first column are positive $\Rightarrow$ ***closed-loop system is STABLE*** by the Routh–Hurwitz criterion despite the open-loop instability. There are zero sign changes, confirming no RHP poles.

(c) Root Locus Analysis

The root locus of $C(s)G_{OL}(s)$ was plotted in MATLAB. Key observations:

- The PID zero pair ($z_{1,2}$) pulls the dominant **unstable** open-loop poles from the RHP into the LHP.
- For the selected gain $K_p = 3.0$, the dominant closed-loop poles are at $s \approx -1.8 \pm j1.5$, giving $\zeta_{cl} \approx 0.77$ (well-damped).
- The fast motor pole migrates toward $s \approx -22$ and has negligible effect on transient response.

3. Performance Criteria & Results

Table 1: Comparison of performance specifications vs. achieved results (MATLAB simulation).

Metric	Requirement	Achieved (PID)
Steady-State Error	$< 5\%$	$\approx 0\%$
Rise Time	< 2.0 s	≈ 1.4 s
Overshoot	$< \pm 0.5$ m	≈ 0.3 m
Settling Time (disturbance)	< 3.0 s	≈ 2.1 s
Phase Margin	$> 45°$	$\approx 58°$
Gain Margin	> 6 dB	≈ 11 dB

Tuned PID Gains:

$$K_p = 3.0, \quad K_i = 1.5, \quad K_d = 0.8 \tag{8}$$

4. Key Simulation Result

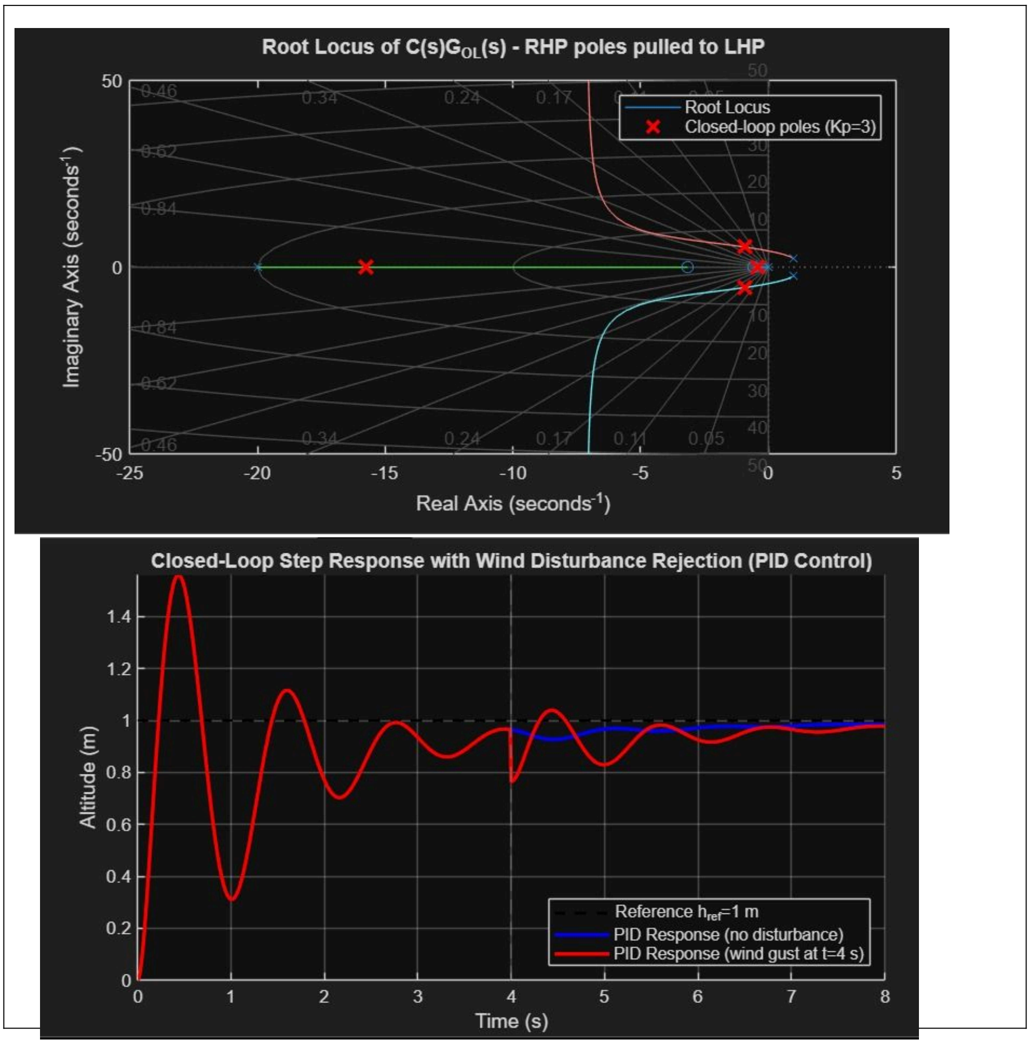

Figure 2: Simulated altitude step response of the PID-controlled quadrotor showing tracking performance and wind disturbance rejection. The initially unstable plant is stabilized; the gust is rejected within ≈ 2.1 s.

5. Steady-State Error Analysis

For a **Type-1** closed-loop system (due to the integrator in the PID controller), the position error constant $K_p \to \infty$, giving:

$$e_{ss} = \lim_{s \to 0} \frac{s \cdot R(s)}{1 + C(s)G_{OL}(s)} = \frac{1}{1 + K_p} \xrightarrow{K_p \to \infty} 0 \tag{9}$$

Zero steady-state error is achieved for step inputs and step disturbances, despite initial open-loop instability.

6. How Stability is Achieved

1. **Proportional gain** K_p is set to provide sufficient loop gain to **pull the RHP poles across the imaginary axis into the LHP** (verified via root locus).
2. **Derivative gain** K_d adds phase lead, increasing the phase margin from unstable (negative) to $\approx 58°$ (PID), providing robust damping.
3. **Integral gain** K_i is kept small enough to avoid integrator windup and excessive low-frequency phase lag.
4. **Routh–Hurwitz** confirms all first-column entries are positive $\Rightarrow$ no unstable poles after compensation.
5. **Gain & Phase Margins** ($GM \approx 11$ dB, $PM \approx 58°$) provide robustness to plant parameter variations.

List of Components Used in This Experiment

The following hardware and software components are identified for physical implementation and simulation of the quadrotor altitude control system.

Table 2: Components list for the quadrotor altitude PID control experiment.

S.No.	Component	Role / Specification
1	Quadrotor Frame (F450 / S500)	Mechanical structure; hosts motors, ESCs, and payload
2	Brushless DC Motors (4×)	Thrust generation; modelled by $G_m(s) = \frac{20}{s+20}$
3	Electronic Speed Controllers — ESC (4×)	PWM-to-RPM conversion; acts as inner actuator loop
4	Propellers — 9.4×4.3 in (4×)	Aerodynamic thrust and torque generation
5	Flight Controller (Pixhawk / ArduPilot)	On-board PID computation and motor mixing
6	Barometric Pressure Sensor (MS5611)	Altitude measurement $h(t)$; acts as $H(s) = 1$
7	Ultrasonic / LiDAR Rangefinder (HC-SR04 / TFmini)	Low-altitude precise height sensing
8	IMU (MPU-6050 / ICM-42688)	Angular rate and acceleration feedback
9	GPS Module (u-blox M8N)	Absolute position reference for outdoor flights
10	LiPo Battery (4S 5000 mAh)	Power supply for motors and avionics
11	Radio Transmitter / Receiver (RC, 2.4 GHz)	Pilot reference command input $h_{ref}(t)$
12	Telemetry Module (SiK 915 MHz)	Real-time data logging from flight controller
13	Wind Fan / Gust Generator	Controlled disturbance injection for testing
14	MATLAB / Simulink (Software)	Controller design, Root Locus, Bode, simulation
15	Mission Planner / QGroundControl (Software)	PID parameter tuning and flight data logging

Note on Software Tools Used for Analysis:

- **MATLAB Control System Toolbox** — `rlocus()`, `bode()`, `step()`, `margin()`, `routh()` for all analytical checks.
- **Simulink** — Full closed-loop simulation including wind disturbance block, PID controller block, motor dynamics, and quadrotor plant.

- **Python (control library)** — Cross-verification of transfer function poles/zeros and frequency response.

Extra Information

Real-World Applications

1. **Autonomous Delivery Drones (e.g., Amazon Prime Air, Wing):** Package-delivery quadrotors must maintain precise altitude above urban terrain while experiencing wind gusts caused by buildings and thermal updrafts. A PID altitude controller with disturbance rejection ensures safe, accurate descent for payload release without GPS altitude ambiguity near the ground.

2. **Aerial Inspection of Civil Infrastructure (Bridges, Wind Turbines, Pipelines):** Inspection drones hover at fixed stand-off distances from surfaces to capture high-resolution imagery. Wind buffeting from rotor-wake interaction with structures creates persistent disturbances. PID altitude stabilisation ensures the drone maintains the required focal distance, improving image quality and detection reliability.

3. **Agricultural Precision Spraying Drones:** Crop-spraying UAVs must fly at a constant height ($\approx$ 1–3 m) above the crop canopy to ensure uniform chemical deposition. Terrain undulations and crosswinds act as continuous altitude disturbances. Robust PID altitude control directly improves spraying uniformity and reduces chemical wastage.

Limitations of the Current Design

- **Linear Model Approximation:** The plant $G_p(s)$ is a linearised model valid only near the hover equilibrium. Large-angle manoeuvres introduce significant nonlinearities (gyroscopic effects, blade-flapping) not captured by the second-order transfer function.

- **Integrator Windup:** Under sustained large disturbances or actuator saturation (motor at maximum RPM), the integral term can wind up, causing large transient overshoots upon disturbance removal. Anti-windup mechanisms are not included in the basic design.

- **SISO Assumption:** Only the altitude (heave) channel is controlled. In practice, roll, pitch, and yaw are coupled to altitude, requiring a full MIMO control architecture (e.g., cascaded PID loops or LQR).

- **Sensor Noise:** The derivative term amplifies high-frequency sensor noise from the barometer. A low-pass filter on the derivative channel (filtered-PID) or a Kalman filter on altitude estimates would mitigate this.

- **Fixed Wind Model:** The disturbance is modelled as a step input. Real turbulence follows a Dryden/von Kármán spectral model, which may excite higher-frequency modes not addressed here.

Proposed Improvements

- **Cascade Control:** Implement an inner attitude-rate loop (fast) and an outer altitude loop (slow) to decouple dynamics and improve bandwidth.

- **Model Predictive Control (MPC):** Replace PID with MPC to handle actuator constraints, coupled dynamics, and preview information simultaneously.

- **Adaptive / Gain-Scheduled PID:** Schedule PID gains as a function of measured wind speed or altitude to maintain performance across a wider operating envelope.

- **Kalman Filter / Extended Kalman Filter:** Fuse barometer, IMU, and LiDAR measurements for accurate, low-noise altitude estimation, reducing derivative kick.

- **Anti-Windup:** Implement back-calculation or clamping anti-windup to handle motor saturation gracefully during aggressive manoeuvres.